

Research Methods

Collect & Analyze Finance Data from LSEG with Python (2025/2026)

Alexandre Garel, Finance Professor, Audencia

Pr. Alexandre Garel



- Professeur de finance durable, Audencia
- Chercheur (climat, biodiversité, AI, politique de vote, finance comportementale)
- Expert dans des comités scientifiques (fonds biodiversité)
- Expert qualifié (FIR – Forum de l'Investissement Responsable)
- Membre du HKUST Global Sustainable Institute
- Missions de conseils (KPMG...)

Pr. Alexandre Garel

Google Scholar



Audencia

Alexandre Garel

MAIN ABOUT PUBLICATIONS WORKING PAPERS PRESS DATA TEACHING

CV

FULL PROFESSOR · FINANCE, AUDENCIA BUSINESS SCHOOL

CLIMATE FINANCE BIODIVERSITY FINANCE ESG BEHAVIORAL FINANCE CORPORATE FINANCE AI

Google Scholar LinkedIn SSRN Email

NAVIGATE RESEARCH & WORK



About

PUBLICATIONS

Firm-Level Nature Dependence [Review of Finance](#)

Mutual Fund Voting on Climate [Mgmt. Science](#)

Employee Diversity & Performance [Fin. Management](#)

Shareholder Engagements in Banks [Econ. Letters](#)

Climate Risk & Mutual Fund Flows [J. Banking & Fin.](#)

Publications

WORKING PAPERS

Climate Disclosure & Director Elections [R&R](#)

Women-Led Mutual Funds Voting [R&R](#)

Climate Risk Engagements

Mandatory Carbon Disclosure

From Vision to Prediction (AI Pipeline)

Working Papers

Material

← ↻ 🔒 https://audencia.blackboard.com/ultra/courses/_48803_1/outline 🔊 ☆ ⋮ Vérifiez qu'il s'agit bien de vous  ...  Conversation

 Cours ▾ Méthodologie de la thèse / Recherche bibliographique • SFIR016_2025  Paramètres du cours **OUVERT**

Contenu Calendrier Annonces Discussions Carnet de notes Messages Analyses Groupes Réussites  Aperçu étudiant

 **RM - Alexandre Garel** ⋮

 Visible pour les étudiants ▾

Python to collect/analyze data from Refinitiv/LSEG Workspace ^

 **COVID_example.ipynb** ⋮

 Visible pour les étudiants ▾

 **dataset_raw_covid_analysis.csv** ⋮

 Visible pour les étudiants ▾

 **cacall_constituents.csv** ⋮

 Visible pour les étudiants ▾

aux messages

[Modifier les paramètres](#)

 Automatisations

[Afficher les automatisations](#)

Détails et actions

 Annuaire

[Afficher tous les membres de votre cours](#)

 Image du cours

[Modifier les paramètres d'affichage](#)

 Class Collaborate ⋮

[Rejoindre la session](#) ▾

 Présence

[Noter la présence](#)

 Manuels et outils

[Afficher les outils du cours et de](#)



How to learn?



1. Refinitiv/LSEG

Install Refinitiv **Workplace**:

<https://cdn.refinitiv.com/Apps/ProductDownloadPage/1.1.13/>



Welcome to **Refinitiv Workplace**

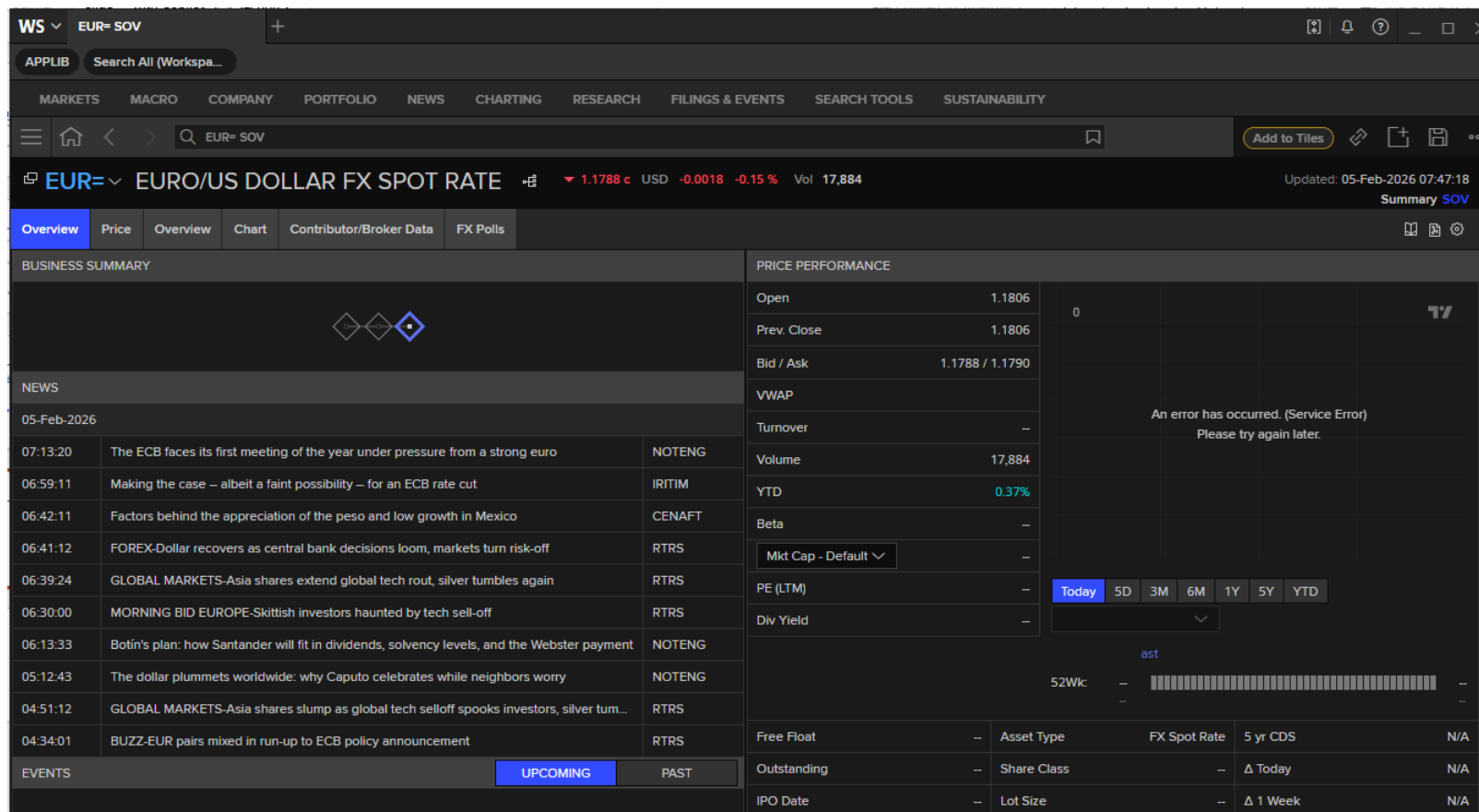
Redesigned for you. With you.

Help us reshape Refinitiv Workplace for anything the future may bring.

Download for Windows

Refinitiv Workplace is also available on [macOS](#) for you to download and install.

1. Refinitiv/LSEG



Knowledge Hub sent you your login and password, shared among several students

Cannot have more than one student using a shared account simultaneously

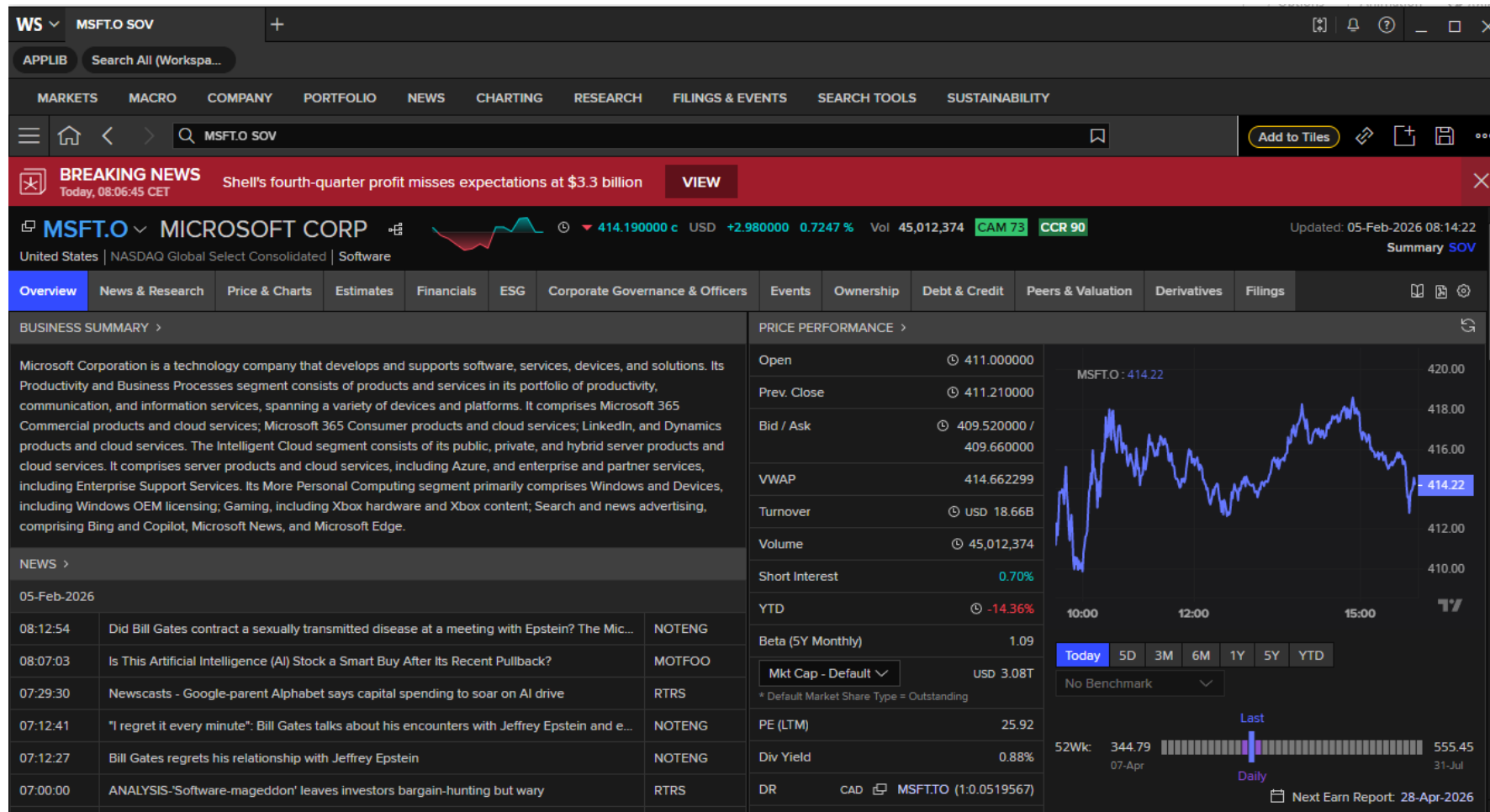
Hence, **collect and save the data** for later running the analyses

1. Refinitiv/LSEG

Available data:

- Accounting (book value, sales, capex, R&D, earnings)
- Financials (bond, stock, option)
- Forecasts (EPS, BPS, price)
- Ownership
- ESG (data points, scores, controversies)
- Filings (annual reports, earning calls, integrated reports)
- Fund information
- Country information
- News flow
- Prices (commodities, FOREX)
- M&A, Deals of PE

1. Refinitiv/LSEG



We search for **Microsoft**

We type Microsoft in the search bar or use its **RIC** (security identifier)

Also works with *ISIN* or *CUSIP*, other security identifiers

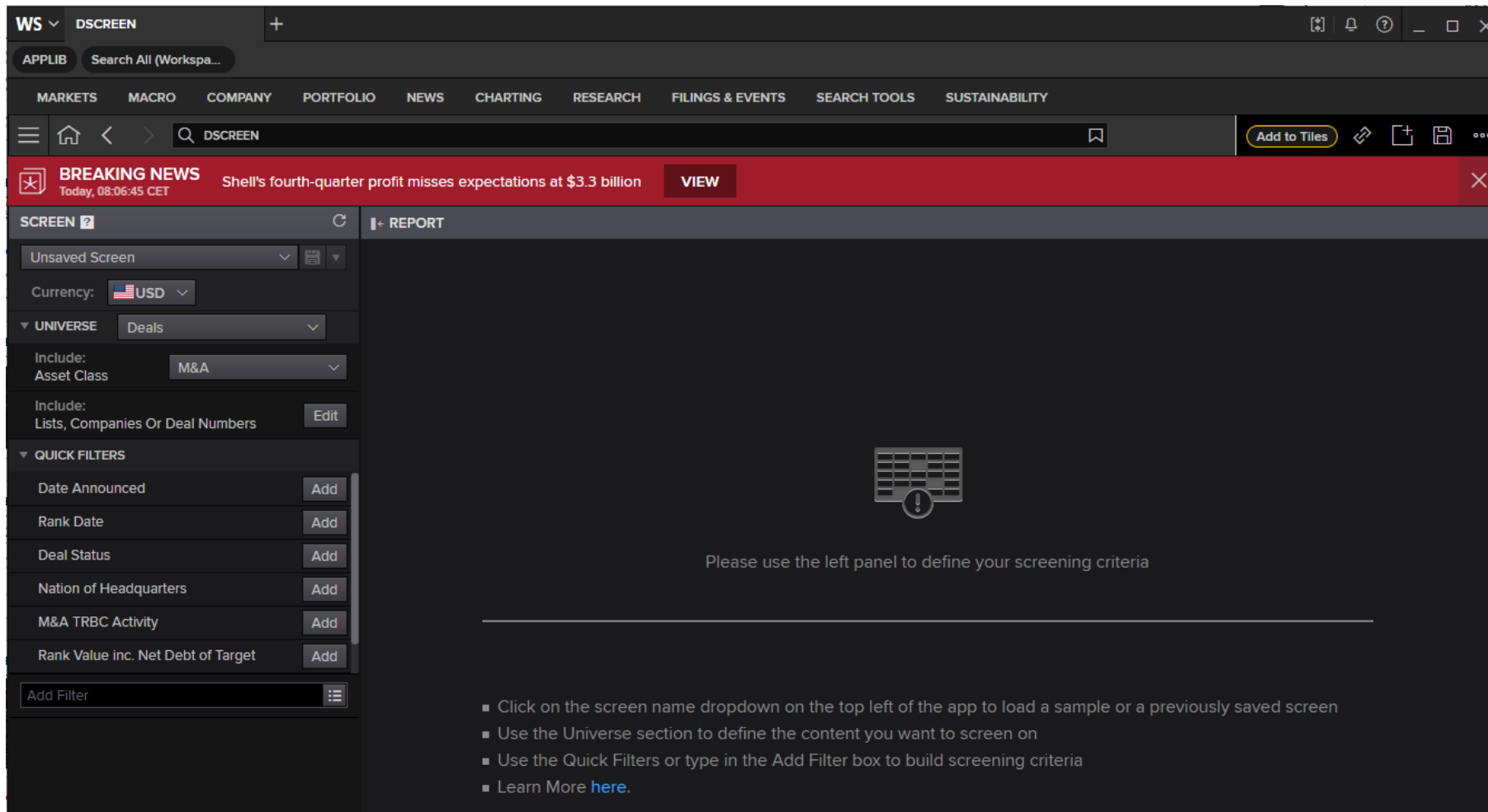
Let us explore the content together

1. Refinitiv/LSEG

Explore:

- Accounting (book value, sales, capex, R&D, earnings)
- Financials (bond, stock, option)
- Forecasts (EPS, BPS, price)
- Ownership
- ESG (data points, scores, controversies)
- Filings (annual reports, earning calls, integrated reports)
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- News flow
- Prices (commodities, FOREX)
- M&A, Deals of PE

1. Refinitiv/LSEG



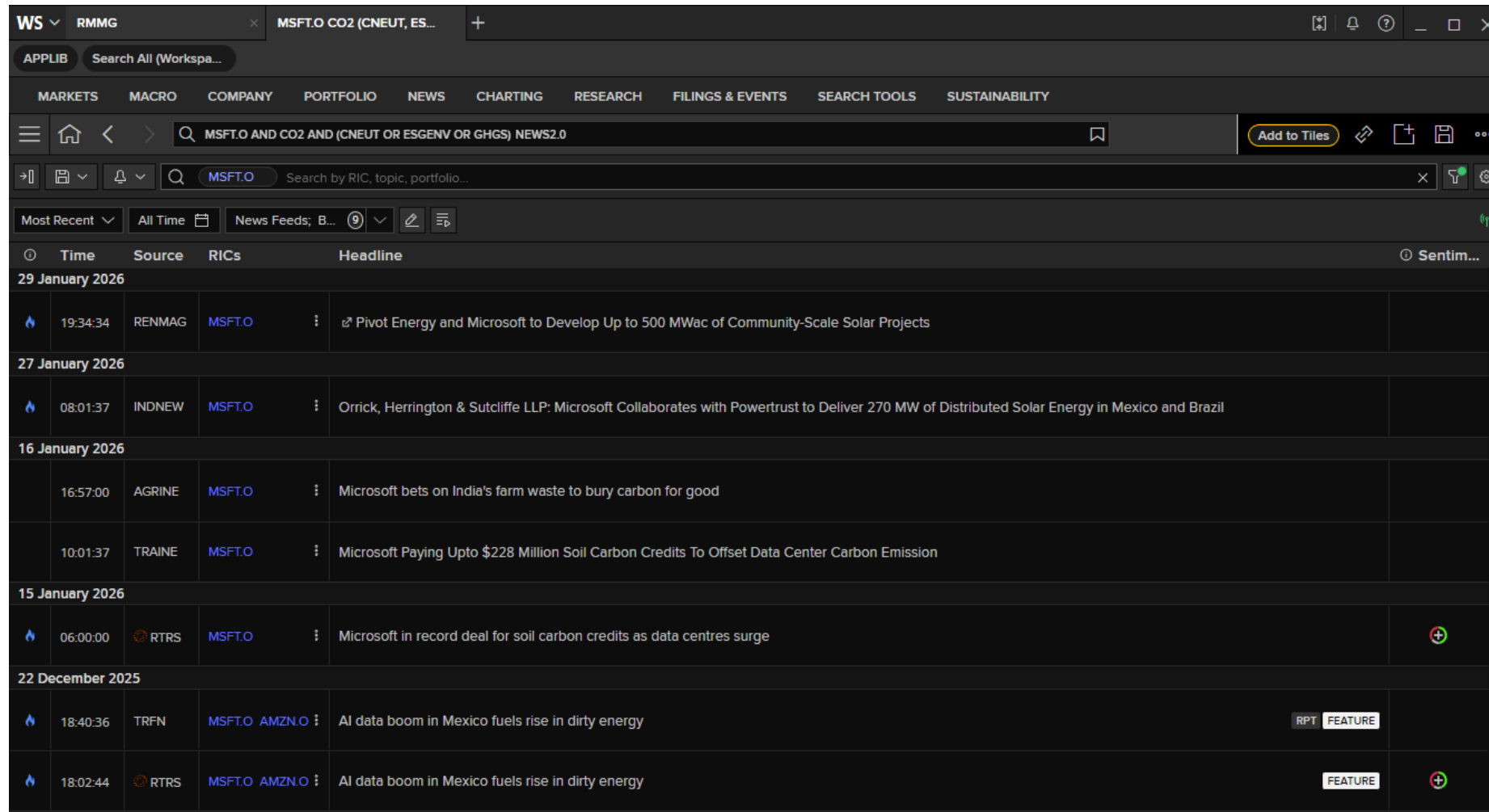
Screensers

Companies

Deals screener (bond,
stock, M&A, PE)

Fund screener

1. Refinitiv/LSEG



The screenshot displays the Refinitiv/LSEG news monitor interface. The top navigation bar includes tabs for APPLIB, Search All (Workspa...), and various market-related categories like MARKETS, MACRO, COMPANY, PORTFOLIO, NEWS, CHARTING, RESEARCH, FILINGS & EVENTS, SEARCH TOOLS, and SUSTAINABILITY. The search bar contains the query "MSFT.O AND CO2 AND (CNEUT OR ESGENV OR GHGS) NEWS2.0". Below the search bar, there are filters for "Most Recent" and "All Time", and a "News Feeds; B..." dropdown. The main table lists news articles with columns for Time, Source, RICs, and Headline. The articles are grouped by date, ranging from 29 January 2026 to 22 December 2025. The table also includes a "Sentim..." column for sentiment analysis.

Time	Source	RICs	Headline	Sentim...
29 January 2026				
19:34:34	RENMA	MSFT.O	↗ Pivot Energy and Microsoft to Develop Up to 500 MWac of Community-Scale Solar Projects	
27 January 2026				
08:01:37	INDNEW	MSFT.O	Orrick, Herrington & Sutcliffe LLP: Microsoft Collaborates with Powertrust to Deliver 270 MW of Distributed Solar Energy in Mexico and Brazil	
16 January 2026				
16:57:00	AGRINE	MSFT.O	Microsoft bets on India's farm waste to bury carbon for good	
10:01:37	TRAINE	MSFT.O	Microsoft Paying Up to \$228 Million Soil Carbon Credits To Offset Data Center Carbon Emission	
15 January 2026				
06:00:00	RTRS	MSFT.O	Microsoft in record deal for soil carbon credits as data centres surge	
22 December 2025				
18:40:36	TRFN	MSFT.O AMZN.O	AI data boom in Mexico fuels rise in dirty energy	RPT FEATURE
18:02:44	RTRS	MSFT.O AMZN.O	AI data boom in Mexico fuels rise in dirty energy	FEATURE

News monitor

1. Refinitiv/LSEG

WS ▾ DIB

APPLIB Search All (Workspa...

MARKETS MACRO COMPANY PORTFOLIO NEWS CHARTING RESEARCH FILINGS & EVENTS SEARCH TOOLS SUSTAINABILITY

⌵ ⌵ < > 🔍 DIB

Add an instrument 🔍 Remove Instrument

🔼 earnings Recently Used All Items ⚙️

Data Item Name	Data Item Code
Earnings	EARNINGS
Earnings	TR.SACTTarEarningsBeforeAnn
Earnings Per Share - Mean	TR.EPSMean
Earnings Per Share Reported - Mean	TR.RepEPSMean
Preferred Earnings - Mean Estimate	TR.PreferredMeasureMeanEst
Earnings Per Share - SmartEstimate®	TR.EpsSmartEst
Earnings Per Share - Predicted Surprise PCT	TR.EpsPreSurprisePct
Price / EPS (SmartEstimate®)	TR.FwdPtoEPSSmartEst
EBIT Margin - %	TR.F.EBITMargPct
EBITDA Margin - %	TR.F.EBITDAMargPct
Income before Taxes to EBIT	TR.F.IncBefTaxtoEBIT
Equity Earnings/(Loss) before Taxes including Non-Recurring	TR.F.EqEarnLossBefTaxInclNonRecur
Normalized Pre-tax Profit	TR.F.NormPretaxProf
Income before Taxes	TR.F.IncBefTax
Net Income after Tax	TR.F.NetIncAfterTax
Equity in Earnings/(Loss) of Affiliates (after Taxes)	TR.F.EqInEarnLossOfAffilAfterTax

Description

Parameters

Earnings

EARNINGS

Definition:
Latest reported earnings per share.
FieldID: 34

Show only: ☒ Cross-Asset Items ☐ Favorites Sort By: ☒ Relevance ☐ A - Z

EARNINGS 📄

Data Item Browser
- DIB -

1. Refinitiv/LSEG

The screenshot displays the Refinitiv CODECR web application interface. At the top, the workspace is named 'WS' and 'CODECR'. A search bar labeled 'Search All (Workspa...)' is present. Below this, a navigation menu includes 'MARKETS', 'MACRO', 'COMPANY', 'PORTFOLIO', 'NEWS', 'CHARTING', 'RESEARCH', 'FILINGS & EVENTS', 'SEARCH TOOLS', and 'SUSTAINABILITY'. The main content area is divided into several sections:

- Instruments:** A search bar with 'MSFT.O' entered.
- DATA ITEMS:** A search bar with 'Search Data Items' and a list of categories on the left, including 'Favorites (0)', 'Real Time (3494)', 'Reference & Identifiers (283)', 'Issuer Level Credit Rating (30)', 'Time Series Ratios (34)', 'Price & Volume (359)', 'Refinitiv I/B/E/S Estimates (19458)', and 'StarMine Models & Analytics (295)'. The 'Fields' section lists various data points like '1 Month Performance to Last Month End', 'Percentage of Fund Assets', 'Holding Name', 'Holding RIC', 'ResourceUse Controversies count', and 'Recent Water and Marine Resource Controversies Count'. A 'Description' panel on the right shows details for 'Number of Shares' (TR.FundNumberOfShares), including its definition and category.
- Selected Items:** A section showing 'TR.FundNumberOfShares' with a 'Drag to Reorder' option and a 'Clear All' button.
- OUTPUT:** A section with a 'Python' tab showing the generated Python code for retrieving the data.

```
import refinitiv.data as rd
rd.open_session()
df = rd.get_data(
    universe = ['MSFT.O'],
    fields = ['TR.FundNumberOfShares']
)
display(df)
```

**Python Code
Generator
- CODECR -**

2. Python – Anaconda - Jupyter

- Why **Python**?
- **Anaconda** bundles Python, Jupyter notebooks, and common data packages in one installer.
 - Installs Python automatically
 - Works on both Mac and Windows
- We download and install it : <http://anaconda.com/download>



2. Python – Anaconda - Jupyter

Anaconda Navigator

File Help

ANACONDA.NAVIGATOR

Update Now Connect

Home

Environments

Learning

Community

All applications on base (root) Channels

 PyCharm The only Python IDE you need – built for data and AI/ML professionals. Supercharged with an AI-enhanced IDE experience. Free forever, plus one month of Pro included. Install	 Anaconda AI Navigator Access various large language models (LLMs) curated by Anaconda, and start leveraging secure local AI today. Install	 Anaconda Toolbox 4.40.0 Anaconda Assistant JupyterLab supercharged with a suite of Anaconda extensions, starting with the Anaconda Assistant AI chatbot. Install	 Anaconda Cloud Notebooks Cloud-hosted notebook service from Anaconda. Launch a preconfigured environment with hundreds of packages and store project files with persistent cloud storage. Launch	 Qt Console 5.6.1 PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more. Launch	 Spyder 6.0.7 Scientific Python Development Environment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features. Launch
 EduBlocks Web-based coding platform from Anaconda designed for students. Learn Python coding through an interactive, block-based visual environment. Launch	 IBM watsonx IBM watsonx is an enterprise-ready AI platform including a data store, model builder, and AI model management and monitoring. Launch	 Oracle Cloud Infrastructure Oracle Data Science Service OCI Data Science offers a machine learning platform to build, train, manage, and deploy your machine learning models on the cloud with your favorite open-source tools. Launch	 PyScript Code and share Python in the Browser. A vibrant community of makers, builders, and hackers building the next frontier of Python-powered web applications. Launch	 PythonAnywhere Host, run, and code Python in the cloud! Get started for free. Launch	 anaconda_powershell_prompt 1.1.0 Opens a PowerShell instance with conda activated (requires menuinst 2.1.1 or greater). Install
 anaconda_prompt 1.1.0 Opens a terminal instance with conda activated (requires menuinst 2.1.1 or greater). Launch	 CMD.exe Prompt 0.1.1 Run a cmd.exe terminal with your current environment from Navigator activated. Launch	 console_shortcut_miniconda 0.1.1 Anaconda Powershell Prompt Launch	 Glueviz 1.4.0 Multidimensional data visualization across files. Explore relationships within and across files. Launch	 JupyterLab 4.5.3 An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and facilitated on the JupyterLab architecture. Launch	 Jupyter Notebook 7.5.0 Web-based, interactive computing notebook environment. Edit and run code, visualize outputs, and describe the workflow. Launch

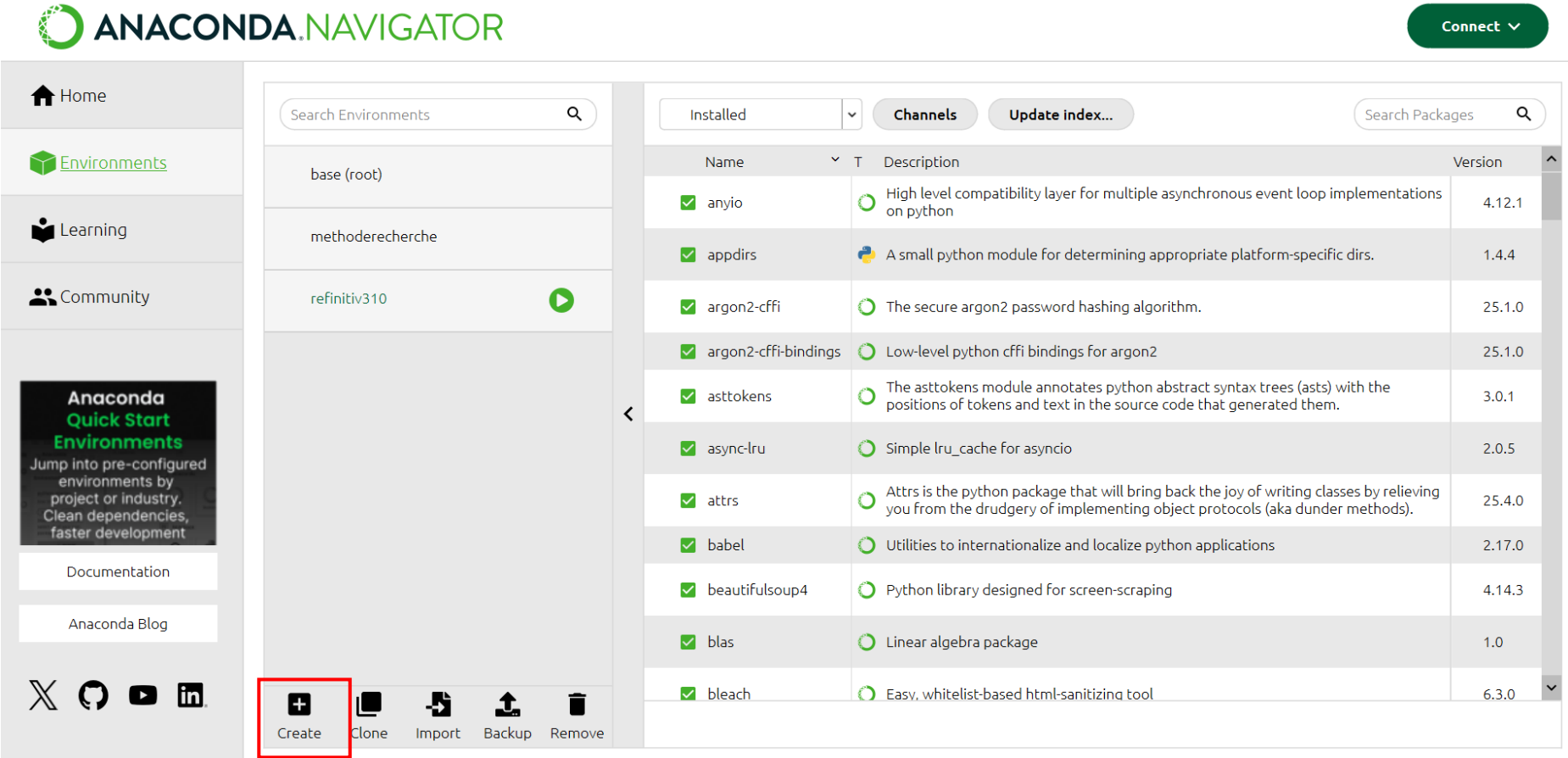
Anaconda Quick Start Environments
Jump into pre-configured environments by project or industry. Clean dependencies, faster development.
[Launch Your Environment](#)

Documentation

Anaconda Blog

2. Python – Anaconda - Jupyter

We create a new environnement, pick a name

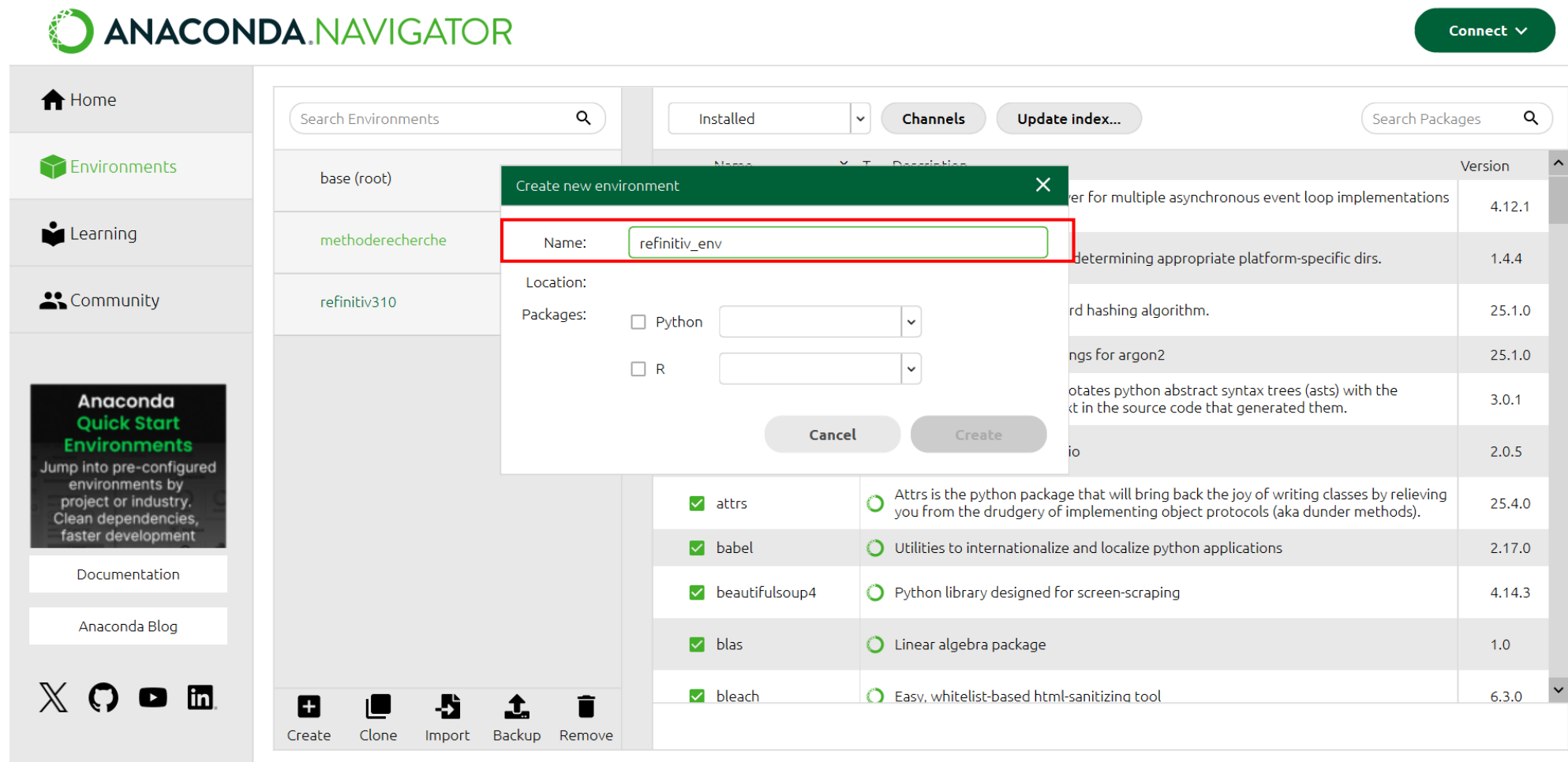


The screenshot displays the Anaconda Navigator application interface. On the left sidebar, the 'Environments' tab is selected. The main panel shows a list of environments: 'base (root)', 'methoderecherche', and 'refinitiv310'. The 'Create' button, represented by a plus icon, is highlighted with a red box at the bottom of the environment list. To the right, the 'Channels' tab is active, displaying a table of installed packages.

Name	Description	Version
anyio	High level compatibility layer for multiple asynchronous event loop implementations on python	4.12.1
appdirs	A small python module for determining appropriate platform-specific dirs.	1.4.4
argon2-cffi	The secure argon2 password hashing algorithm.	25.1.0
argon2-cffi-bindings	Low-level python cffi bindings for argon2	25.1.0
asttokens	The asttokens module annotates python abstract syntax trees (asts) with the positions of tokens and text in the source code that generated them.	3.0.1
async-lru	Simple lru_cache for asyncio	2.0.5
attrs	Attrs is the python package that will bring back the joy of writing classes by relieving you from the drudgery of implementing object protocols (aka dunder methods).	25.4.0
babel	Utilities to internationalize and localize python applications	2.17.0
beautifulsoup4	Python library designed for screen-scraping	4.14.3
blas	Linear algebra package	1.0
bleach	Easy, whitelist-based html-sanitizing tool	6.3.0

2. Python – Anaconda - Jupyter

We create a new environnement, pick a name



ANACONDA NAVIGATOR Connect

Home Environments Learning Community

Search Environments Installed Channels Update index... Search Packages

base (root) methoderecherche refinitiv310

Create new environment

Name:

Location:

Packages: ☒ Python ☒ R

Cancel Create

Name	Description	Version
attrs	Attrs is the python package that will bring back the joy of writing classes by relieving you from the drudgery of implementing object protocols (aka dunder methods).	25.4.0
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Create Clone Import Backup Remove

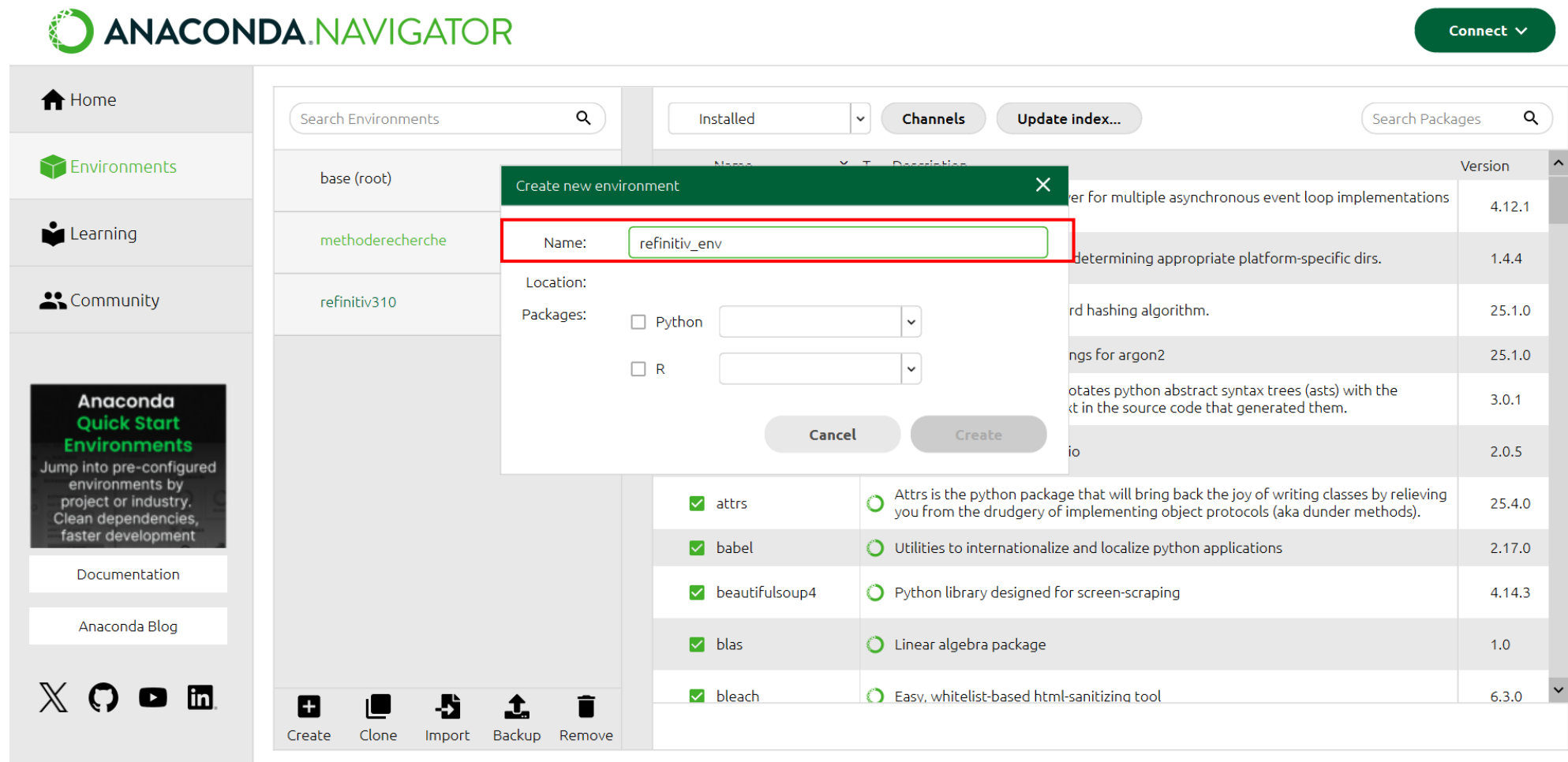
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Documentation Anaconda Blog

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2. Python – Anaconda - Jupyter

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ANACONDA NAVIGATOR Connect

Home Environments Learning Community

Search Environments Installed Channels Update index... Search Packages

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Create Clone Import Backup Remove

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2. Python – Anaconda - Jupyter

We use **Python 3.9** – otherwise there are issues with the Refinitiv package

The screenshot displays the Anaconda Navigator application interface. On the left sidebar, there are navigation options: Home, Environments, Learning, and Community. The main panel shows a list of environments: 'base (root)', 'methoderecherche', and 'refinitiv310'. A 'Create new environment' dialog box is open in the center, with the following details:

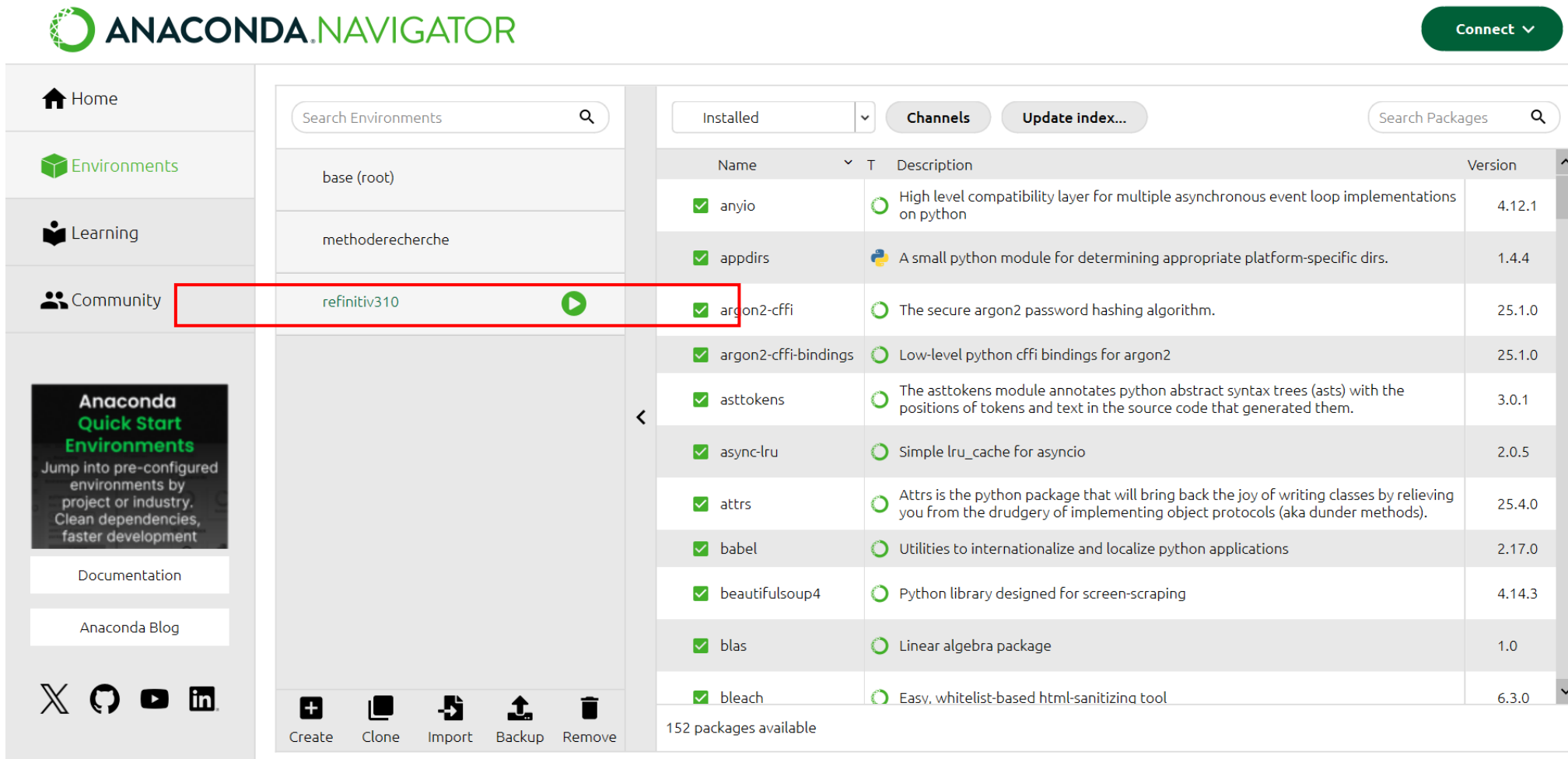
- Name: refinitiv_env
- Location: (empty)
- Packages: ☒ Python 3.9.25 (highlighted with a red box)
- ☐ R 3.6.1

At the bottom of the dialog are 'Cancel' and 'Create' buttons. The background interface includes a 'Search Packages' bar and a table of installed packages:

Name	Version
attrs	25.4.0
babel	2.17.0
beautifulsoup4	4.14.3
blas	1.0
bleach	6.3.0

2. Python – Anaconda - Jupyter

Select the new environment ('refinitiv310' in my case) and go home to *install Jupyter*



The screenshot shows the Anaconda Navigator application interface. The left sidebar contains navigation links: Home, Environments, Learning, and Community. The 'Environments' tab is active, showing a list of environments: 'base (root)', 'methoderecherche', and 'refinitiv310'. The 'refinitiv310' environment is highlighted with a red box. Below the environment list, there are buttons for 'Create', 'Clone', 'Import', 'Backup', and 'Remove'. The main panel displays the 'Channels' tab for the selected environment, showing a list of installed packages. The 'Update index...' button is visible. The bottom status bar indicates '152 packages available'.

ANACONDA NAVIGATOR Connect

Home Environments Learning Community

Search Environments

base (root)
methoderecherche
refinitiv310

Installed Channels Update index... Search Packages

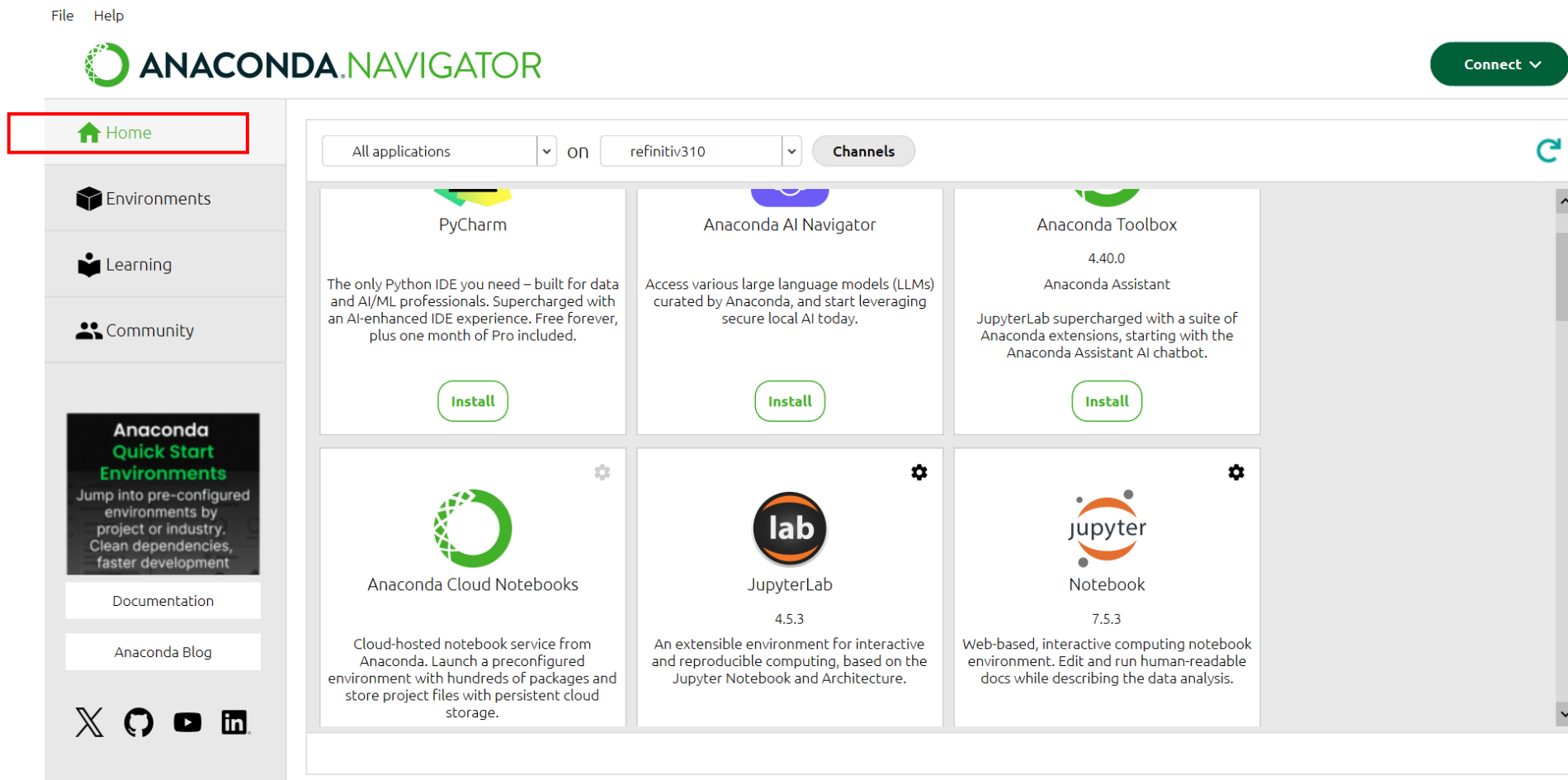
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152 packages available

Create Clone Import Backup Remove

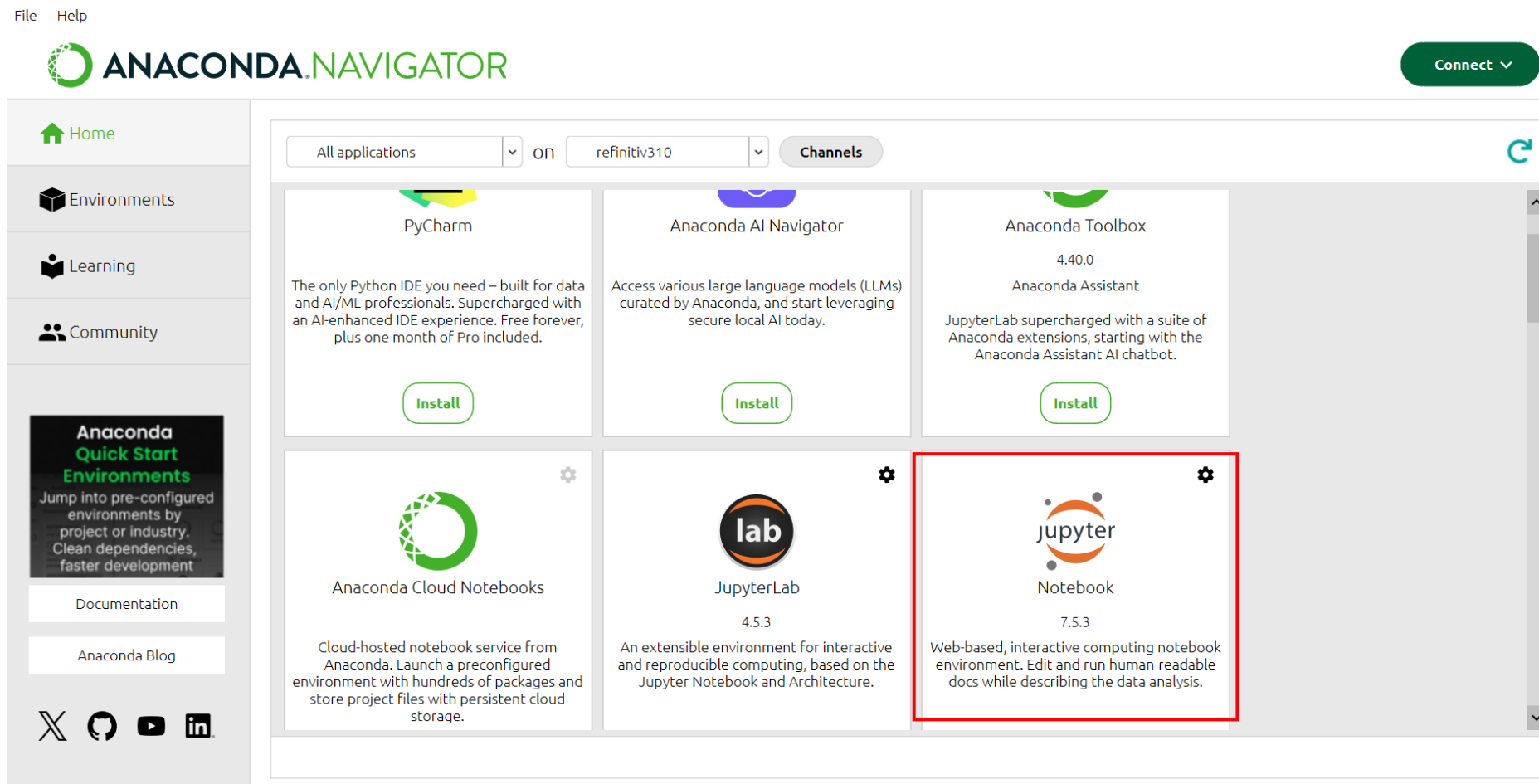
2. Python – Anaconda - Jupyter

Select the new environment ('refinitiv310' in my case) and go home to *install Jupyter*



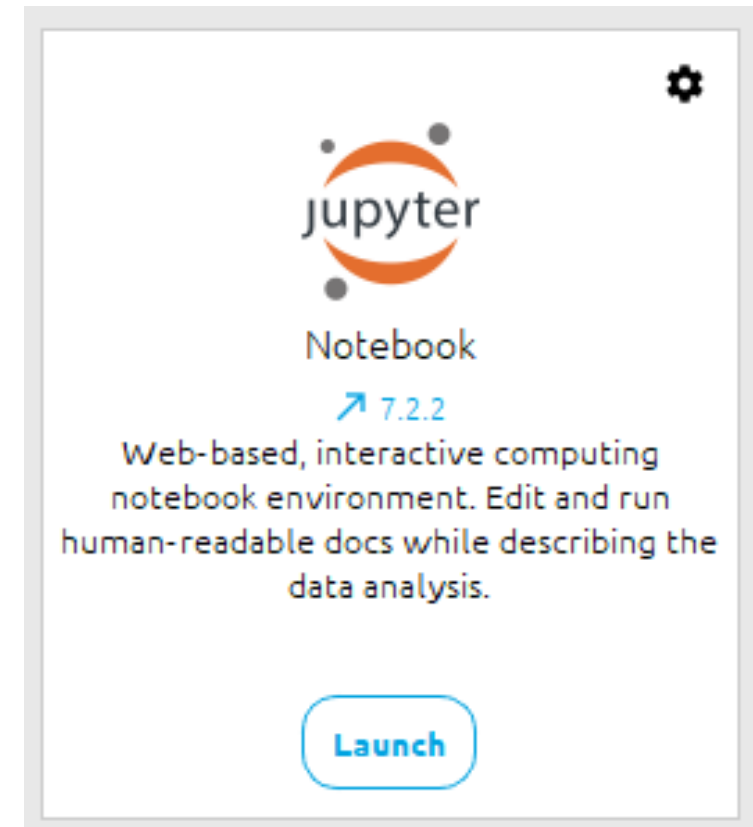
2. Python – Anaconda - Jupyter

Select the new environment ('refinitiv310' in my case) and go home to *install Jupyter*



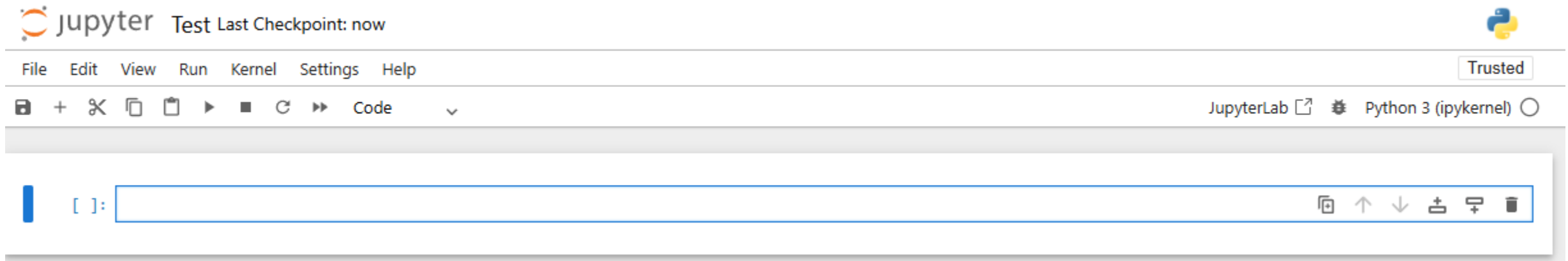
2. Python – Anaconda - Jupyter

- Install/launch **Jupyter Notebook**
- Jupyter Notebook is a tool that lets you write code, run it, and see the results all in one place—inside your web browser.
- Allows to run code step by step/ see output immediately (numbers, tables, plots) / add text explanations next to the code.



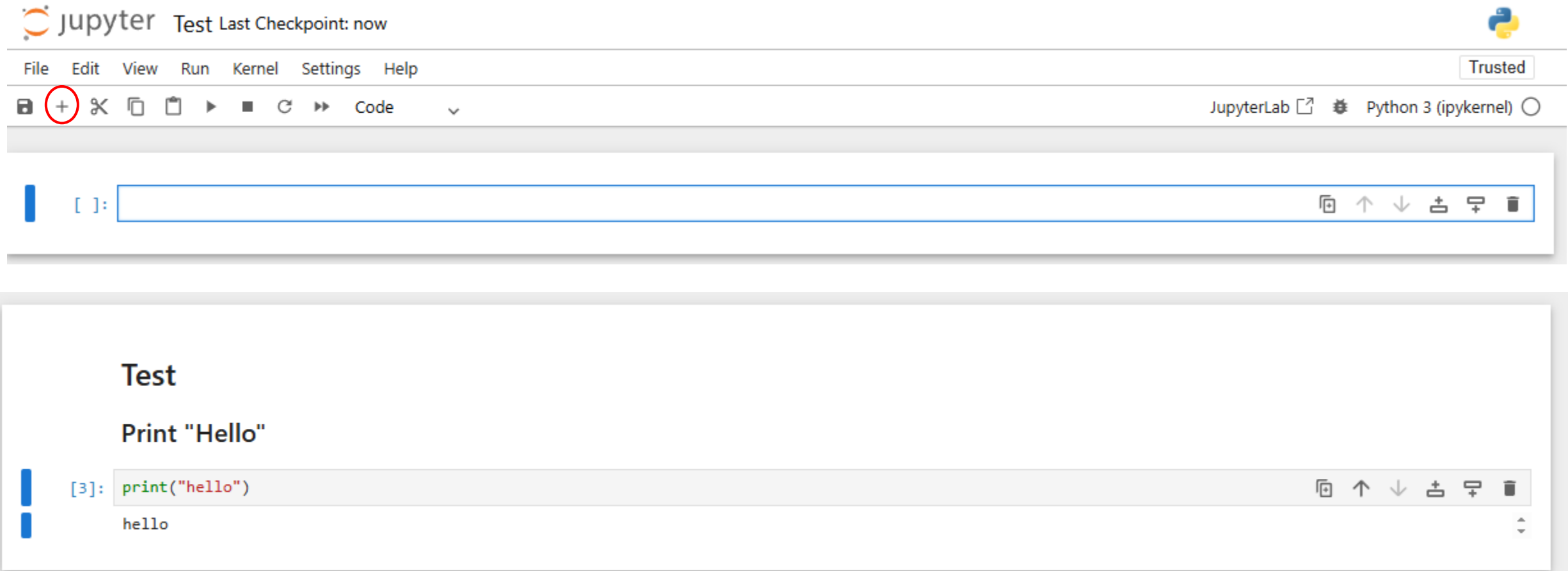
2. Python – Anaconda - Jupyter

- Launch Jupyter Notebook
- Choose folder
- File > New > Notebook (Python 3)
- Save it as *Test.ipynb*



2. Python – Anaconda - Jupyter

➤ With the + button, we can add text (markdown) or code section (python), and then execute



The screenshot displays the JupyterLab web interface. At the top, the 'jupyter' logo is on the left, and the text 'Test Last Checkpoint: now' is next to it. On the right, there is a Python logo and a 'Trusted' status box. Below this is a menu bar with 'File', 'Edit', 'View', 'Run', 'Kernel', 'Settings', and 'Help'. A toolbar follows, containing icons for saving, adding a new cell (highlighted with a red circle), deleting, copying, pasting, running, and other actions. The 'Code' dropdown menu is open. On the right side of the toolbar, it says 'JupyterLab' with an external link icon, a settings gear icon, and 'Python 3 (ipykernel)' with a refresh icon. The main workspace shows a single code cell with the prompt '[]:' and an empty text input field. Below this, a second code cell is shown with the prompt '[3]:' and the code `print("hello")`. The output of this cell is 'hello'. The top of this second cell has the heading 'Test' and the sub-heading 'Print "Hello"'.

2. Python – Anaconda - Jupyter

We switch to the [first_tests.ipynb](#)

All the material, including this presentation is available on **Blackboard**.

3. Collecting data with Python

We switch to the [research_method_notebook.ipynb](#)

All the material, including this presentation is available on **Blackboard**.

3. Collecting data with Python

Now that we have reviewed the part of the **research_method_notebook** dedicated to data collection, try to collect datasets possibly relevant for your own master thesis...

4. Collecting and analyzing data with Python

We switch to the **COVID_example.ipynb**

All the material, including this presentation is available on **Blackboard**.